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Computational Logic Handouts

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Contents

Chapter 1
LoI

1.1 Well formed formulas

Exercise 1.1 Which of the following symbols are used in LoI?

1. $\sqcap$	6. $\sqsubseteq$	11. $\oplus$
2. $\neg$	7. $\supset$	12. $\forall$
3. $\vee$	8. $\wedge$	13. $\Diamond$
4. $\equiv$	9. $\models$	14. $\Box$
5. $\sqcup$	10. $\exists$	15. $\otimes$

Exercise 1.2 Determine which of the following is a wff in LoI.

1. $a \wedge b$
2. $A(b) \supset B(c)$
3. $\exists \alpha.(\alpha = a)$
4. $\forall A(\alpha).(B(\alpha))$
5. $\forall \alpha.\exists \beta.((A(\alpha) \oplus B(\beta)) \supset \exists \gamma.(\alpha = \gamma \wedge \beta \neq \gamma))$

1.2 Translations

Exercise 1.3 Indicate which of the formalizations in LoI of the following natural language sentences are correct (one or more):

1. The formalization of "Stray cats are agile" in LoI is:

   $\forall x.((Cat(x) \wedge Stray(x)) \supset Agile(x))$

2. The formalization of "Red cats are not smart" in LoI is:

   $\forall x.((\neg Smart(x) \wedge Cat(x)) \supset Red(x))$

3. The formalization of "If a cat is of at least three different colors, then it is female" in LoI is:

   $\forall c.\exists x.\exists y.\exists z.((color(c,x) \wedge color(c,y) \wedge color(c,z) \wedge Cat(c)) \supset Female(c))$

4. The formalization of "There is a cat that cannot hunt because it is domesticated" in LoI is:

   $\exists x.((Cat(x) \wedge Domesticated(x)) \supset \neg canHunt(x))$

5. The formalization of "There are at most two friendly cats" in LoI is:

   $\forall x.\forall y.((Friendly(x) \vee Friendly(y)) \supset (x = y))$

Exercise 1.4 Indicate which of the formalizations in LoI of the following natural language sentences are correct (one or more):

1. The formalization of "The floating balloons contain helium" in LoI is

   $\neg\exists x.(Balloons(x) \wedge Floating(x) \wedge \neg contain(x, helium))$

2. The formalization of "There is a red balloon that does not float" in LoI is

   $\exists x.(((Baloon(x) \wedge isRed(x)) \supset \neg Floating(x))$

3. The formalization of "If there are at least two balloons, then at least one is red" in LoI is

   $\exists x \exists y.(Baloon(x) \wedge Baloon(y) \wedge x \neq y) \supset \exists z.(Baloon(z) \wedge isRed(z))$

4. The formalization of "There are at most two red balloons that do not float" in LoI is

   $\forall x \forall y.((B(x) \wedge B(y) \wedge isR(x) \wedge isR(y) \wedge \neg F(x) \wedge \neg F(y)) \supset (x = y \vee y = x)$

5. The formalization of "There is exactly one red balloon" in LoI is

   $\exists x.(B(x) \wedge R(x)) \wedge \forall y \forall z.((B(y) \wedge R(y) \wedge B(z) \wedge R(z)) \supset (z = y))$

Exercise 1.5 Indicate which of the formalizations in LoI of the following natural language sentences are correct (one or more):

1. The formalization of "If a car is fast, then it is a race car" in LoI is:

   $\forall x.((Car(x) \wedge Fast(x)) \supset Race(x))$

2. The formalization of "Fast cars are not cheap" in LoI is:

   $\forall x.((\neg Cheap(x) \wedge Car(x)) \supset Fast(x))$

3. The formalization of "If a car has more than two seats, then it is not a race car" in LoI is:

   $\forall m.\exists x.\exists y.\exists z.((seat(m, x) \wedge seat(m, y) \wedge seat(m, z) \wedge Car(m)) \supset \neg Race(m))$

4. The formalization of "There is a car that is cheap because it is used" in LoI is:

   $\exists x.((Car(x) \wedge Used(x)) \supset Cheap(x))$

5. The formalization of "There is at most one steering wheel in each car" in LoI is:

   $\forall m.\forall x.\forall y.((Car(m) \wedge (SteeringWheel(m, x) \vee SteeringWheel(m, y)) \supset (x = y))$

6. The formalization of "There is at most one steering wheel in each car" in LoI is:

   $\forall m.\forall x.\forall y.((Car(m) \wedge (SteeringWheel(m, x) \wedge SteeringWheel(m, y)) \supset (x = y))$

Exercise 1.6 Indicate which of the formalizations in LoI of the following natural language sentences are correct (one or more):

1. The formalization of "Savory pancakes do not contain sugar" in LoI is:

   $\forall x.(SavoryPancake(x) \wedge \neg contains(x, sugar))$

2. The formalization of "There is a non-savory pancake that does not contain sugar" is:

   $\exists x.((Pancake(x) \wedge \neg Savory(x)) \supset \neg contains(x, sugar))$

3. The formalization of “If there are at least two pancakes, then at least one is savory” is:

$$\exists x \exists y.(Pancake(x) \wedge Pancake(y) \wedge (x \neq y)) \supset \exists z.Pancake(z) \wedge Savory(z))$$

4. The formalization of “There are at most two unsavory pancakes” is

$$\forall x \forall y.((P(x) \wedge P(y) \wedge \neg S(x) \wedge \neg S(y)) \supset (x = y \vee y = x)$$

5. The formalization of “There is exactly one sweet pancake” in LoI is:

$$\exists x.(P(x) \wedge D(x)) \wedge \forall y \forall z.((P(y) \wedge D(y) \wedge P(z) \wedge D(z)) \supset (z = y))$$

Exercise 1.7 Formalize the following sentence in LoI: “There exists only one person who goes to the mountains and loves skiing” given the following mapping:

- P(x) -> ”x is a person”
- M(x) -> ”x goes to the mountains”
- S(x) -> ”x loves skiing”

Exercise 1.8 Formalize the following sentence in LoI: “Only one student passed the exam without studying” given the following mapping:

- St(x) -> “x is a student”
- P(x) -> “x passed the exam”
- Sd(x) -> “x studied”

Exercise 1.9 Translate the following LoDE theory in LoI.

$City \sqsubseteq \exists name.String$
$Person \sqsubseteq \exists bornIn.City$
$Book \sqsubseteq \exists title.String \sqcap \exists topic.String$
$Philosophy \sqsubseteq \exists mainBelief.String \sqcap Discipline$
$Citizen \equiv Person \sqcap \exists address.String \sqcap \exists livesIn.City$
$Philosopher \equiv Person \sqcap \exists expertIn.Philosophy \sqcap \forall wrote.Book$

$Citizen(Bill\#1)$
$Person(Sara\#1)$
$City(Trento\#1)$

Exercise 1.10 Translate the following LoDE theory in LoI.

$RestEmployee \equiv Person \sqcap \forall worksAt.Restaurant$
$RestOwner \equiv Person \sqcap \forall ownsA.Restaurant$
$Cook \equiv RestaurantEmployee \sqcap \exists canPrepare.Recipe$
$Waiter \equiv RestaurantEmployee \sqcap \exists serve.Table$
$Chef \equiv Cook \sqcap \exists headOf.Kitchen$

$Chef(Mike\#1)$
$RestEmployee(Will\#1)$
$RestOwner(Robert\#1)$
$ownsA(Robert\#1, R\#1)$

Exercise 1.11 Translate the following LoDE theory in LoI.

$Employee \equiv Person \sqcap \exists worksAt.ShopPlace$
$ShopPlace \equiv Place \sqcap \exists sells.Thing$
$Bartender \equiv Employee \sqcap \forall worksAt.Bar$
$Bar \equiv ShopPlace \sqcap \forall sells.Drink$

$Bartender(E\#1)$
$Bar(S\#1)$
$Drink(T\#1)$

1.3 Reasoning problems

Exercise 1.12 Given the domain D, determine which of the listed interpretations are a model for the theory T.

$D = \{1, 2, 3, 4, 5, 6\}$

$$T = \begin{cases} \forall x.(P(x) \supset \neg Q(x)) \\ \exists x.Q(x) \\ \forall x.\exists y.(r(x, y) \supset Z(y)) \end{cases}$$

1.
$$I_1 = \begin{cases} P = \{1, 2, 3\} \\ Q = \{4, 5, 6\} \\ Z = \{3, 4\} \\ r = \{(1, 3), (4, 4)\} \end{cases}$$

2.
$$I_2 = \begin{cases} P = \{1, 2, 3, 4, 5\} \\ Q = \{4, 5, 6\} \\ Z = \{3, 4\} \\ r = \{(1, 3)\} \end{cases}$$

3.
$$I_3 = \begin{cases} P = \{1\} \\ Q = \{4, 5, 6\} \\ Z = \{3, 4, 5, 6\} \\ r = \{\} \end{cases}$$

4.
$$I_4 = \begin{cases} P = \{1, 2, 3\} \\ Q = \{4, 5, 6\} \\ Z = \{3, 4\} \\ r = \{(1, 3), (4, 2)\} \end{cases}$$

5.
$$I_5 = \begin{cases} P = \{1, 2, 3, 4\} \\ Q = \{\} \\ Z = \{3, 4, 1, 2\} \\ r = \{(1, 3), (4, 2), (2, 4)\} \end{cases}$$

6.
$$I_6 = \begin{cases} P = \{\} \\ Q = \{1, 2, 3, 4, 5, 6\} \\ Z = \{3, 4\} \\ r = \{(1, 3), (4, 4)\} \end{cases}$$

Exercise 1.13 Given the domain D, determine which of the listed interpretations are a model for the theory T.

$D = \{1, 2, 3, 4, 5, 6\}$

$$T = \begin{cases} \forall x.(P(x) \oplus Q(\mathcal{F}(x))) \\ \forall x.(\mathcal{F}(x) \neq a) \end{cases}$$

1.
$$I_1 = \begin{cases} a = 1 \\ P = \{1, 2, 3\} \\ Q = \{4, 5, 6\} \\ \mathcal{F}(n) = \begin{cases} 6 & if\ n = 6 \\ n + 1 & otherwise \end{cases} \end{cases}$$

2.
$$I_2 = \begin{cases} a = 4 \\ P = \{1, 2, 3\} \\ Q = \{4, 5, 6\} \\ \mathcal{F}(n) = \begin{cases} n & if\ n \in P \\ n + 1 & if\ n \in Q \wedge n \neq 6 \\ 6 & if\ n = 6 \end{cases} \end{cases}$$

3.
$$I_3 = \begin{cases} a = 1 \\ P = \{1\} \\ Q = \{3, 4, 5, 6\} \\ \mathcal{F}(n) = \begin{cases} 6 & if\ n = 6 \\ n + 1 & otherwise \end{cases} \end{cases}$$

4.
$$I_4 = \begin{cases} a = 1 \\ P = \{1\} \\ Q = \{3, 4, 5, 6\} \\ \mathcal{F}(n) = \begin{cases} 6 & if\ n = 1 \\ n - 1 & otherwise \end{cases} \end{cases}$$

5.
$$I_5 = \begin{cases} a = 6 \\ P = \{1\} \\ Q = \{3, 4, 5, 6\} \\ \mathcal{F}(n) = \begin{cases} 2 & if\ n = 1 \\ n - 1 & otherwise \end{cases} \end{cases}$$

6.
$$I_6 = \begin{cases} a = 3 \\ P = \{1, 2, 3, 4, 5, 6\} \\ Q = \{2, 4\} \\ \mathcal{F}(n) = \begin{cases} 2 & if\ n = 3 \\ 4 & otherwise \end{cases} \end{cases}$$

1.4 Solutions

Solution **1.1** $\neg \vee \equiv \supset \wedge \models \oplus \exists \forall$

Solution **1.2**

1. False
2. True
3. True, 'a' can be a constant, and = is a valid operator
4. False
5. True

Solution **1.3**

1. True
2. False. It excludes the existence of non-smart cats that are not red.
3. False. It does not exclude the possibility that one of the colors may be repeated.
4. True
5. False. It should be "and" instead of "or."

Solution **1.4**

1. True, formula to be inverted
2. False, implication weakens the existential quantifier
3. True
4. False, a variable is missing to enclose "at most two"
5. True

Solution **1.5**

1. True
2. False
3. False
4. True
5. False
6. True

Solution **1.6**

1. False
2. False
3. True
4. False
5. True

Solution **1.7** The formalization must express the conjunction of two true facts, the first stating that a person exists, the second stating that there is only one. There are different formalizations. We have indicated some of them below.
$\exists x.(P(x) \wedge S(x) \wedge M(x)) \wedge \forall y.\forall z.((P(y) \wedge P(z) \wedge G(y) \wedge G(z) \wedge S(y) \wedge S(z)) \supset (y = z))$
$\exists x.(P(x) \wedge S(x) \wedge M(x) \wedge \forall y.((P(y) \wedge S(y) \wedge M(y))) \supset (y = x)))$

Solution **1.8** Same as previous exercise.

Solution **1.9**

$$
\begin{aligned}
&\forall x.((City(x) \supset (\exists y.(name(x, y) \wedge String(y))))\\
&\wedge (Person(x) \supset (\exists y.(bornIn(x, y) \wedge City(y))))\\
&\wedge (Book(x) \supset (\exists y.(title(x, y) \wedge String(y)) \wedge \exists y.(topic(x, y) \wedge String(y))))\\
&\wedge (Philosophy(x) \supset (Discipline(x) \wedge \exists y.(mainBelief(x, y) \wedge String(y))))\\
&\wedge (Citizen(x) \equiv (Person(x) \wedge \exists y.(address(x, y) \wedge String(y)) \wedge \exists y.(livesIn(x, y) \wedge City(y))))\\
&\wedge (Philosopher(x) \equiv (Person(x) \wedge \exists y.(expertIn(x, y) \wedge Philosophy(y))\\
&\qquad \wedge \forall y.(wrote(x, y) \supset Book(y)))))\\
&Citizen(Bill\#1) \wedge Person(Sara\#1) \wedge City(Trento\#1)
\end{aligned}
$$

Solution **1.10** Apply the same procedure as the previous exercise.

Solution **1.11** Apply the same procedure as the previous exercise.

Solution **1.12**

1. True
2. False
3. True
4. False
5. False
6. True

Solution **1.13**

1. False
2. True
3. True
4. False
5. False
6. False